Bulacan Polytechnic College
Malolos, Bulacan

CT-MAD-213 MOBILE APPLICATION DEVELOPMENT
FINAL PROJECT DOCUMENTATION

BPC CAMPUS NAVIGATOR

A Mobile Campus Map and Information Application

Category: Maps and Geolocation Applications

Submitted by:
GROUP 3
Millet Cabrera
Mary Grace Zuniga
Charlotte Galuyo
Marvin Garcia
Jenylyn Banag

ACT - 2B

Submitted to:
Mr. Kyle Joseph Fabian

Date: October 24, 2026
Version 1.0

1. PROJECT OVERVIEW

1.1 Project Description

BPC Campus Navigator is a mobile application that helps students, visitors, and staff of Bulacan Polytechnic College find buildings, offices, and facilities on campus. It combines an interactive campus map, a route guide, a directory of offices, class schedules, announcements, and campus resources in a single app. Authorized administrators can add, edit, and delete the content of the app directly from an Admin Dashboard.

1.2 Problem Statement

New students, transferees, parents, and visitors often have difficulty locating buildings, offices, and rooms on the campus. Information such as enrollment schedules, office hours, and announcements is scattered across bulletin boards and social media posts, and is not always updated. This results in lost time, confusion during enrollment periods, and repeated questions to staff and security personnel.

1.3 Target Users

- Students (new, transferee, and continuing) who need to locate classrooms, offices, and facilities.
- Visitors and parents who are unfamiliar with the campus.
- Faculty and staff who need quick access to schedules and announcements.
- Administrators who maintain the map, announcements, offices, schedules, and resources.

2. OBJECTIVES, SCOPE AND FEATURES

2.1 General Objective

To design and develop a mobile application that helps the Bulacan Polytechnic College community navigate the campus and access campus information quickly and accurately.

2.2 Specific Objectives

- To provide an interactive campus map with search and category filters for buildings and offices.
- To provide a route guide that directs users to a chosen destination.
- To provide announcements, a directory of offices, and campus resources in one application.
- To provide an Admin Dashboard for managing map locations, announcements, offices, schedules, and resources (CRUD).

2.3 Scope and Limitations

Scope:

- Interactive campus map with 19 sample locations, search bar, and category chips.
- Home, Map, Schedule, Announcements, and Profile screens with bottom navigation.
- Sign In / Sign Up, Edit Profile, and Sign Out.
- Admin Dashboard with Map, Announcements, Offices, Schedule, and Resources tabs.

Limitations:

- The map covers the Bulacan Polytechnic College campus only.
- Class schedules are available only to signed-in users.
- Users who sign up have view-only access. Only administrators can add, edit, or delete map locations, announcements, offices, schedules, and resources.

- The app does not use a database.
- The application does not use real-time GPS tracking. Users navigate the campus using the custom campus map, location pins, search function, and available route feature.

3. SYSTEM DESIGN

3.1 Application Flow Diagram

Figure 1 shows how a user moves through the application. After launch, the user lands on the Home screen and uses the bottom navigation bar to reach Map, Schedule, Announcements, and Profile. Schedule requires sign-in, and the Admin Dashboard is available only to administrators.

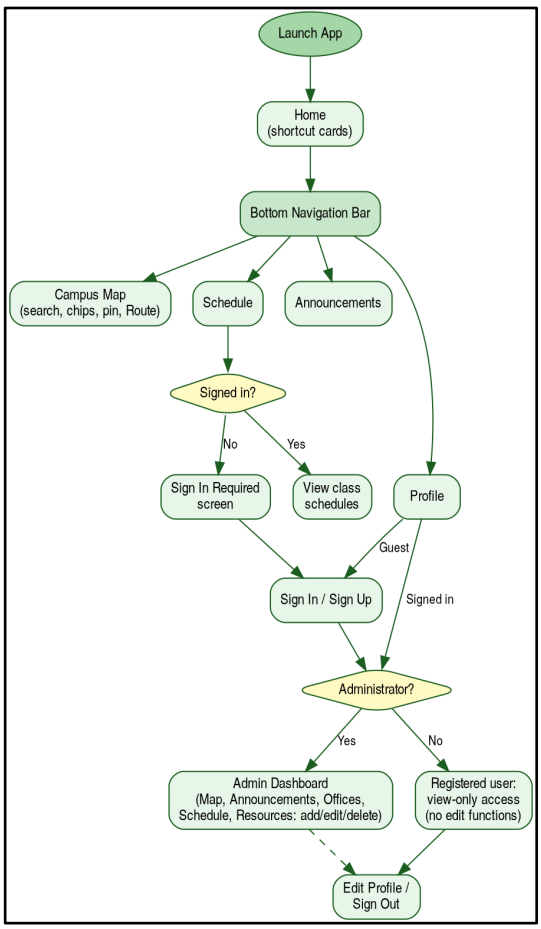


Figure 1. Application flow diagram

Architecture note: The application is developed using Kotlin and Java. It uses local storage and does not use a database. Add the actual IDE and framework details if required.

3.2 Use Case Diagram

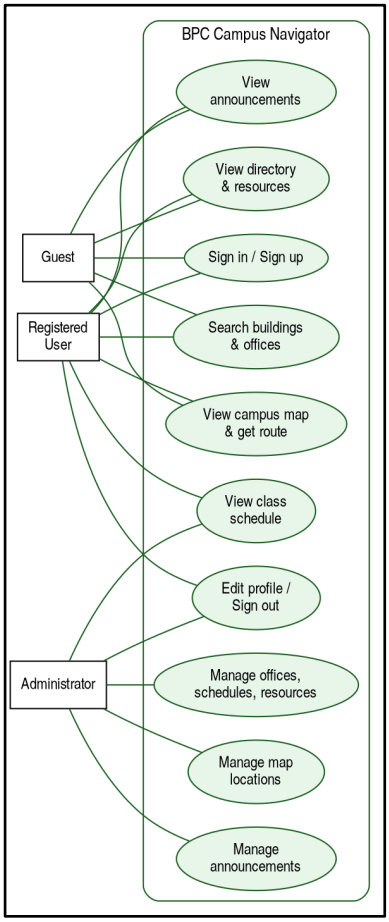


Figure 2. Use case diagram for Guest, Registered User, and Administrator

3.3 ER Diagram and Data Storage

An ER diagram is required only if a database is used. BPC Campus Navigator does not use a database, so an ER diagram is not applicable. The app data (map locations, announcements, offices, class schedules, resources, and user accounts) is handled within the application itself. The application uses local storage and does not use a database. Specify the exact local storage method and where user accounts are stored after verifying the implementation.

4. UI/UX AND IMPLEMENTATION

4.1 UI/UX Design

The interface follows a consistent green theme that reflects the college seal. Every screen uses the same header (screen title, short subtitle, and BPC logo), white rounded cards on a light background, and the same button styles. The bottom navigation bar keeps the five main sections reachable in one tap and highlights the current screen.

Element	Design Decision
Color palette	Dark green (#1B5E20 family) for headers and primary buttons; medium green for accents and active states; white cards; light green background; red used only for delete and sign-out actions.
Typography	Condensed display font with bold headings and smaller body text.
Navigation	Bottom tab bar (Home, Map, Schedule, Announcements, Profile); admin sections use a top tab bar inside the Admin Dashboard.

Element	Design Decision
Consistency	Same header, card style, icon set, and button shapes across all screens.

4.1.1 Wireframes

The wireframes below show the layout of the main screens before visual styling: a header with the screen title and logo, the main content area, and the bottom navigation bar. Boxes with a cross are placeholders for images or maps.

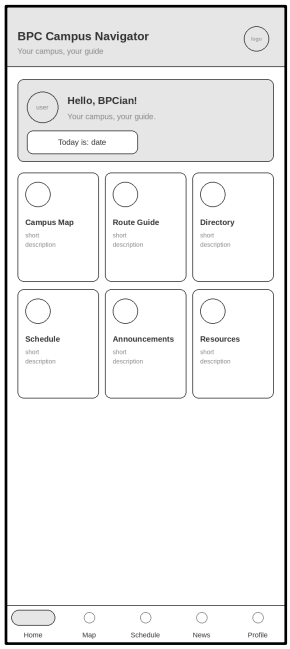


Figure 3. Wireframe: Home

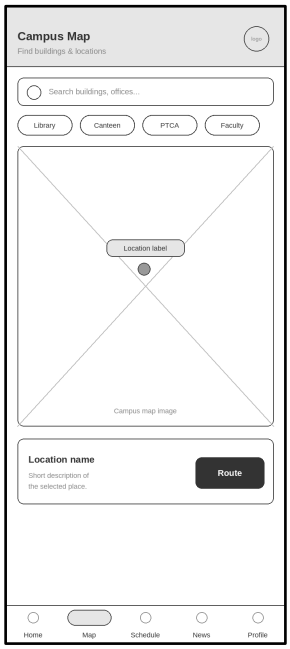


Figure 4. Wireframe: Campus Map

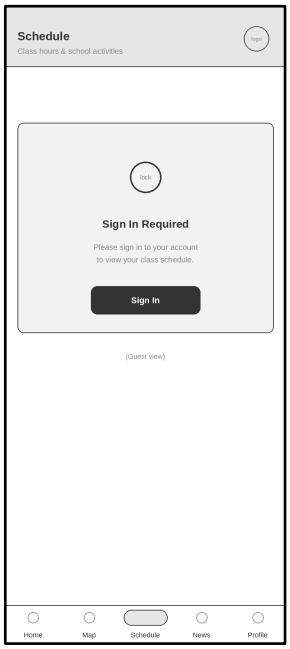


Figure 5. Wireframe: Schedule (guest)

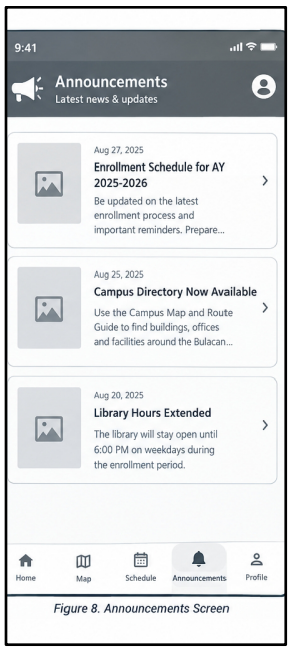


Figure 8. Announcements Screen

Figure 6. Wireframe: Announcements



Figure 7. Wireframe: Profile (administrator)



Figure 8. Wireframe: Sign In

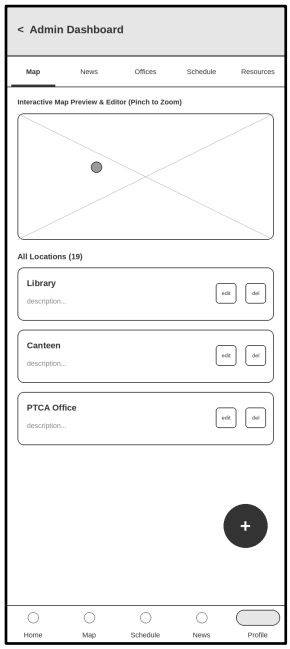


Figure 9. Wireframe: Admin Dashboard

4.2 Development Tools and Technologies

Category	Tool / Technology
Programming language / framework	Kotlin and Java
IDE / editor	. Android Studio,
Database / backend	No database used. Data is handled inside the app. Local storage within the application; no database is used.
Map implementation	Custom campus map image with tappable pins and pinch-to-zoom (as shown in the Admin map editor).
Design tools	Canva
Version control	Git and GitHub
Testing devices	Android smartphone

4.3 Major Application Screens

The following figures show the main screens of the application.

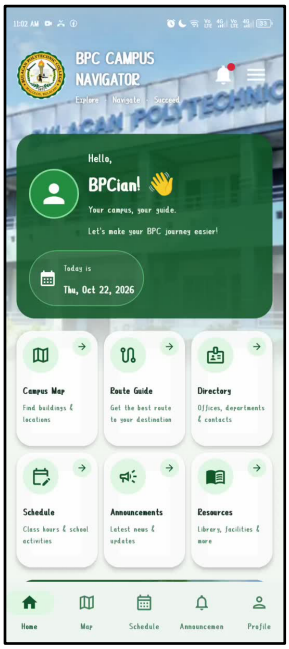


Figure 10. Home screen

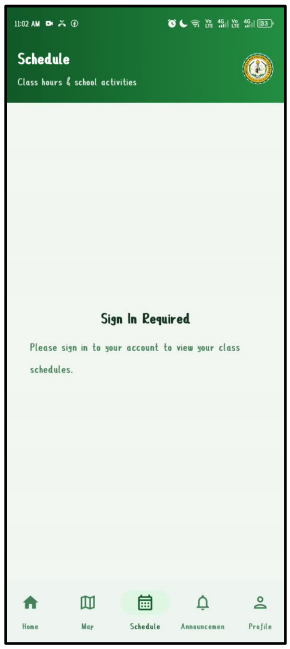


Figure 11. Schedule (guest)

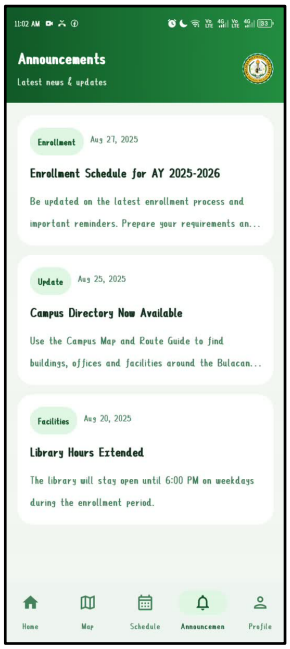


Figure 12. Announcements

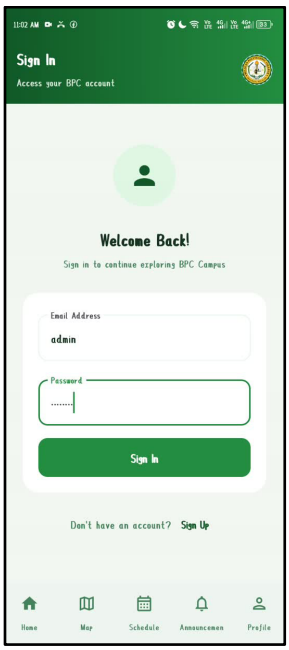


Figure 13. Sign In



Figure 14. Campus Map with pin

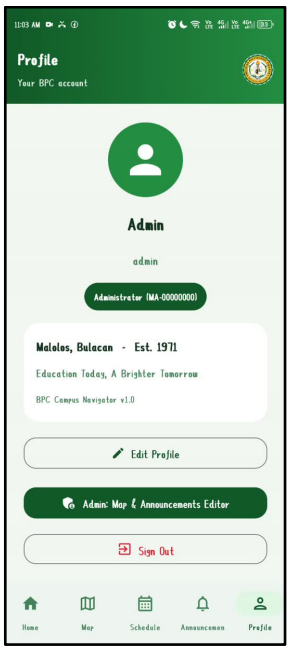


Figure 15. Profile (Administrator)

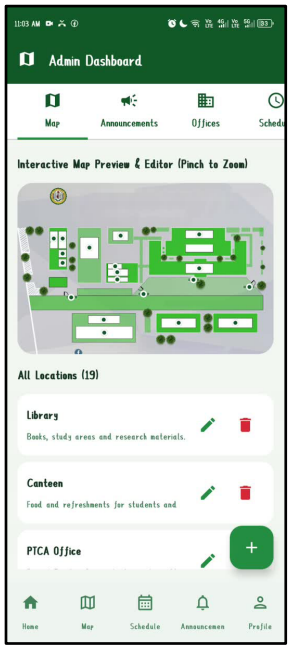


Figure 16. Admin Dashboard: Map

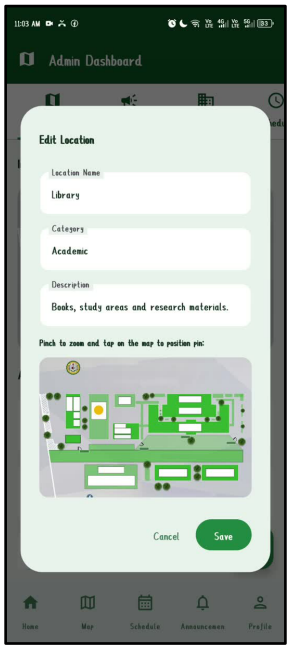


Figure 17. Edit Location dialog

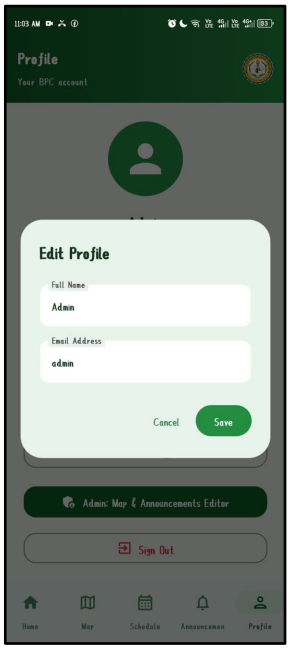


Figure 18. Edit Profile dialog

4.4 Feature Implementation

5. TESTING AND EVALUATIONy

5.1 Test Cases and Results

Tests marked "Passed" were performed and observed in the recorded demonstration of the application. Tests marked "Failed" did not produce the expected result and are discussed in Section 5.2 and Section 6.2.

ID	Test Case	Expected Result	Result	Remarks
TC-01	Open the app	Home screen shows greeting, date, and six shortcut cards	Passed	Observed
TC-02	Open Schedule as guest	"Sign In Required" message is shown	Passed	Observed
TC-03	Sign in with valid administrator account	Profile shows administrator role and admin button	Passed	Observed
TC-04	Sign in with wrong password	The application rejects invalid credentials and displays an error message.	Failed	The application allows sign-in despite invalid credentials.
TC-05	Tap map category chips	Pin and label move to selected location; details card updates	Passed	Observed
TC-06	Select a building using map pins	Matching location is shown on the map	Passed	The selected location and its details are displayed when the user taps a map pin.
TC-07	Tap Route on a location	Route to the destination is displayed	Passed	Observed

ID	Test Case	Expected Result	Result	Remarks
TC-08	Open Admin Dashboard tabs	Map, Announcements, Offices, Schedule, and Resources lists load	Passed	Observed
TC-09	Open Edit Location dialog	Fields show current values with mini map for pin	Passed	Observed
TC-10	Add, edit, and delete a record as administrator	The administrator can add, edit, and delete records successfully.	Passed	Observed
TC-11	Sign out	App returns to Guest view	Passed	Observed
TC-12	Sign up as a regular user and sign in	The registered user account remains available after the application is closed and reopened.	Failed	Registered user accounts are not properly saved or displayed for the administrator.
TC-13	Open app without internet / on other devices	The application opens without an internet connection.	Passed	The application opens successfully without an internet connection.

5.2 Testing Results Summary

A total of 13 test cases were executed to evaluate the functionality and reliability of the BPC Campus Navigator application. The results were categorized as Passed or Failed based on the observed application behavior: 11 test cases passed and 2 failed (TC-04 and TC-12). TC-04 showed that the Sign In feature accepted invalid credentials, and TC-12 showed that newly registered user accounts were no longer available after the application was closed and reopened. These issues need to be addressed and the application re-tested.

6. CONCLUSION AND RECOMMENDATIONS

6.1 Project Accomplishments

- Developed a working campus navigation app with more than three functional screens and working navigation.
- Implemented an interactive map with search, category chips, pins, and a Route button.
- Implemented user roles with protected schedules and an Admin Dashboard with full add, edit, and delete functions.
- Applied a consistent, easy-to-use green interface across all screens.

6.2 Problems Encountered

- Insufficient Input Validation in Sign Up

  One of the problems encountered during the development of BPC Campus Navigator was insufficient input validation in the Sign Up feature. The application allowed users to proceed with registration even when the entered information was incorrect or incomplete. This could result in invalid account information and affect the reliability of the registration process. Input validation is recommended to ensure that required fields are completed and that the entered information follows the expected format before registration is allowed.

- Registered Users Are Not Visible to the Administrator
  One of the problems encountered during the development of the BPC Campus Navigator was that newly registered users could not be viewed by the administrator. Although users could create accounts, the system did not have a feature that allowed the administrator to view the list of registered users. This limitation affected the administrator's ability to monitor registered accounts. Implementing a user management feature is recommended for future development.
- Insufficient Authentication and Input Validation
  During testing, the application allowed users to sign in using a name instead of a valid email address and an arbitrary password.

This indicates that the Sign In feature does not properly validate the email format or verify the user's credentials. This issue may allow unauthorized access to the application. The authentication process needs to be improved to ensure that only users with valid registered accounts can sign in

- Registered User Accounts Are Not Persistently Saved
  During testing, the team discovered that newly registered user accounts did not persist after the application was closed and reopened. As a result, registered users could not reliably access their accounts after restarting the application. This issue may be related to the current account storage and authentication implementation. Improving persistent account storage and authentication is recommended to ensure that registered user accounts remain available when the application is reopened.

6.3 Recommendations / Future Improvements

- Add real-time GPS location and turn-by-turn route display on the campus map.
- Add push notifications for new announcements and class schedule reminders.
- Add a database or cloud backend so administrator updates are shared with all users.
- Add an offline mode so that the map and directory work without internet.
- Add a user management feature that allows the administrator to view and manage registered user accounts securely.
- Integrate GPS functionality in the future to help users identify their current location on campus and improve navigation to selected destinations
- Improve the authentication system by validating email addresses and verifying passwords against registered accounts. The application should reject invalid credentials and display an appropriate error message.
- Improve the Sign Up validation by checking required fields, validating email formats, and ensuring that passwords meet the application's requirements before allowing registration.
- Implement persistent storage or a secure backend authentication system to ensure that registered user accounts remain available after the application is closed and reopened.